Numerator: Ordered Linearly Independent k-tuples in $\mathbb{F}_q^n$ (an $n \times k$ matrix)

$$\begin{pmatrix} v_{11} & v_{12} & \cdots & v_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nk} \end{pmatrix} \times \begin{pmatrix} v_{11} & v_{12} & \cdots & v_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nk} \end{pmatrix} \times \cdots \times \begin{pmatrix} v_{11} & v_{12} & \cdots & v_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nk} \end{pmatrix}$$

$\mathbf{v}_1 \neq \mathbf{0}$ — $q^n - 1$ ways

$\mathbf{v}_2 \notin \mathrm{Span}(\mathbf{v}_1)$ — $q^n - q$ ways

$\mathbf{v}_k \notin \mathrm{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$ — $q^n - q^{k-1}$ ways

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{\text{Numerator}}{\text{Denominator}}$$

Denominator: Ordered Bases for a fixed k-dimensional subspace (a $k \times k$ matrix)

$$\begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ u_{k1} & u_{k2} & \cdots & u_{kk} \end{pmatrix} \times \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ u_{k1} & u_{k2} & \cdots & u_{kk} \end{pmatrix} \times \cdots \times \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ u_{k1} & u_{k2} & \cdots & u_{kk} \end{pmatrix}$$

$\mathbf{u}_1 \neq \mathbf{0}$ — $q^k - 1$ ways

$\mathbf{u}_2 \notin \mathrm{Span}(\mathbf{u}_1)$ — $q^k - q$ ways

$\mathbf{u}_k \notin \mathrm{Span}(\mathbf{u}_1, \ldots, \mathbf{u}_{k-1})$ — $q^k - q^{k-1}$ ways